Hospital Operations & Patient Analytics

Project Storyboard | Tableau Dashboard Narrative

Project purpose

This project organizes patient admission information and hospital infrastructure information into interactive Tableau dashboards. The dashboards are designed to help a viewer explore admission patterns, patient flow, medical-condition patterns, and reported infrastructure measures.

Data used

Two datasets are used: a Patient Admissions dataset and a PMC Hospital Infrastructure dataset. They are treated as separate sources in the dashboards. The storyboard does not assume a common facility identifier or a joined dataset.

Storyboard sequence

Scene 01: Define the operational questions

Hospital operations involve several related questions: how many admission records are present, how admissions change over time, how long patients stay, which admission types are recorded, and what infrastructure capacity and footfall are reported.

Purpose in the story: Establishes the questions the dashboards are intended to help explore.

Scene 02: Prepare the data

The project uses the prepared Patient Admissions and PMC Hospital Infrastructure datasets. Patient admission measures are drawn from the patient dataset, while bed capacity and monthly footfall measures are drawn from the infrastructure dataset.

Purpose in the story: Keeps measures tied to their correct source and avoids implying an unsupported join.

Scene 03: Hospital Overview

The first dashboard presents five headline KPIs: Total Admissions, Average Length of Stay, Total Billing Amount, Average Patient Age, and Emergency Admissions. Supporting charts show Monthly Admissions Trend, Admissions by Medical Condition, and Admissions by Type.

Purpose in the story: Gives the viewer a high-level summary before exploring individual operational patterns.

Scene 04: Patient Flow

The second dashboard focuses on the movement of admission records over time. It combines Total Admissions, Average Length of Stay, and Emergency Admissions with a monthly admissions trend, a length-of-stay distribution, and admissions by type.

Purpose in the story: Helps the viewer examine timing, stay-duration distribution, and the mix of admission types.

Scene 05: Medical Condition Analysis

The third dashboard compares medical conditions using admissions by condition, average length of stay by condition, and total billing by condition. It also displays Total Admissions, Average Length of Stay, and Total Billing Amount as KPI cards.

Purpose in the story: Allows comparison of condition categories across volume, stay duration, and recorded billing.

Scene 06: Resource Utilization

The fourth dashboard uses the PMC infrastructure source. It displays Total Hospital Beds and Average Monthly Patient Footfall, alongside beds by Type (Hospital / Facility) and average monthly footfall by Type (Hospital / Facility).

Purpose in the story: Summarizes reported bed capacity and footfall across the categories supplied in the infrastructure data.

Scene 07: Interpret findings and limitations

The dashboards provide descriptive views of the supplied records. They can help identify patterns and categories for further review, but they do not by themselves establish causes, service quality, staffing adequacy, profitability, or real-time operational status.

Purpose in the story: Encourages responsible interpretation and follow-up analysis.

Key measures and interpretation

The following notes describe how the displayed measures should be understood based on the fields and dashboard configurations shown.

Measure	Meaning and interpretation
Total Admissions	Counts admission records using the Name field. It should not be described as distinct patients because a verified unique patient ID has not been established.
Average Length of Stay	Average of Length of Stay (Days) in the Patient Admissions dataset.
Total Billing Amount	Sum of Billing Amount in the Patient Admissions dataset. Recorded billing is not necessarily cash collected, revenue recognized, or profit.
Average Patient Age	Average of the Age field in the Patient Admissions dataset.
Emergency Admissions	Count of admission records classified as Emergency under Admission Type.
Total Hospital Beds	Sum of Number of Beds in facility type in the PMC infrastructure source. Represents reported capacity, not occupied or available beds.
Average Monthly Patient Footfall	Average monthly footfall measure reported in the PMC infrastructure source.

Expected project value

Together, the four dashboards provide a structured descriptive view of admission activity, patient flow, medical-condition patterns, and reported infrastructure measures. Hospital stakeholders can use the views to identify areas that may warrant closer investigation, ask more focused operational questions, and plan additional data collection where important measures are unavailable.

Limitations to keep in mind

The patient and infrastructure sources are separate. Medical Condition is not a department field. Bed totals do not measure occupancy or availability. Footfall does not by itself measure bed utilization. Findings are descriptive and should not be presented as causal conclusions.

Prepared from the four Tableau dashboard screenshots and project details shared in this conversation. This storyboard describes the displayed dashboards and their intended reading sequence; it does not claim additional findings beyond the available fields.